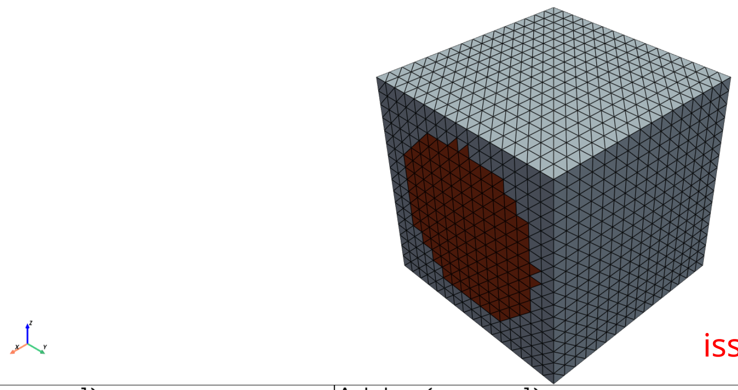
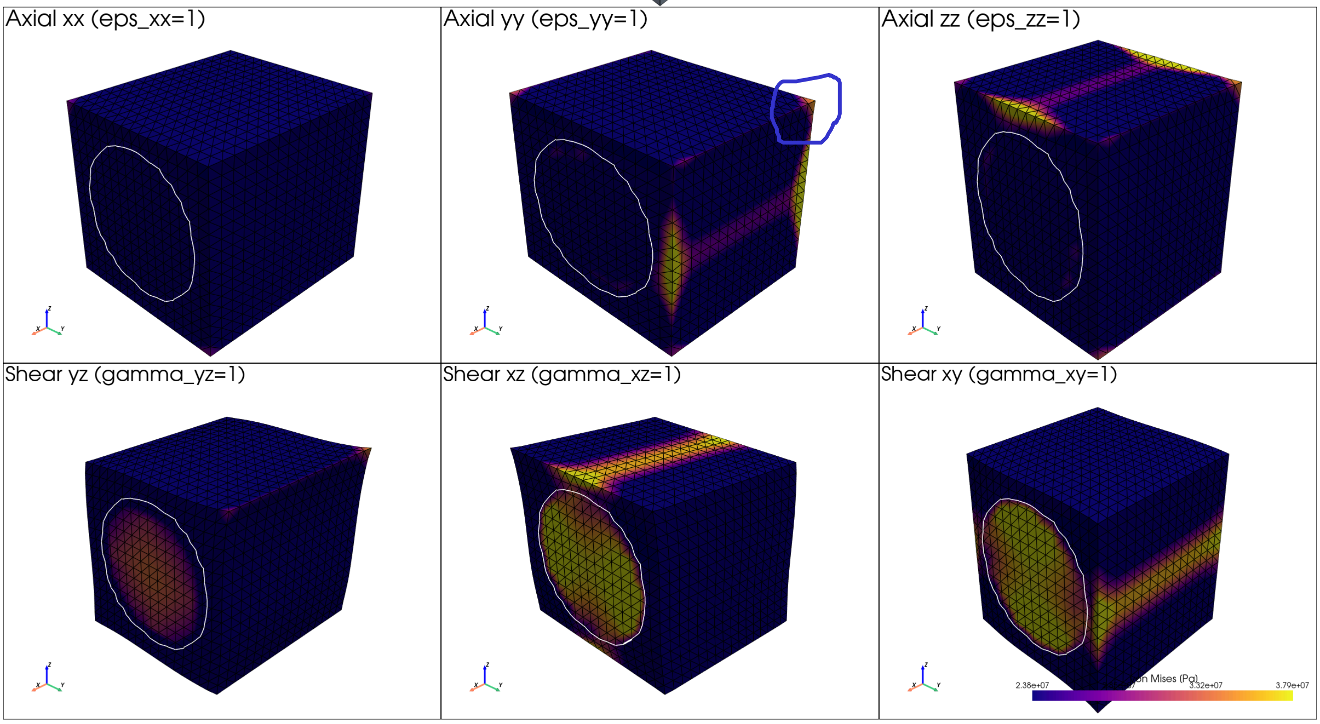
/home/wr1/projects/b3/b3_tex/results/uniaxial_deformation_iso.png

FE mesh (tet) — 16x16x16 structured + 24576 cells; yarn = red, matrix = blue (cell-centroid phase classification)



issue persists



INPUTS

material / geometry	values
matrix	E_L=3.0 GPa, E_T=3.0 GPa, G_LT=1.11 GPa, nu_LT=0.35, nu_TT=0.35
yarn	E_L=140.0 GPa, E_T=10.0 GPa, G_LT=5.00 GPa, nu_LT=0.28, nu_TT=0.40
yarn cylinder	r=0.4, vf=0.503, axis=(1.0,0.0,0.0)
mesh	(16, 16, 16) (tetrahedron)

OUTPUTS (effective stiffness)

effective constant	value
E_x, E_y, E_z [GPa]	72.0, 5.9, 5.9
nu_xy, nu_xz, nu_yz	0.311, 0.311, 0.434
G_xy, G_xz, G_yz [GPa]	2.25, 2.25, 1.79
max diag [GPa]	74.12

FULL C_eff [GPa] (Voigt order 11,22,33,23,13,12)

	11	22	33	23	13	12
11	74.12	3.34	3.34	−0.00	−0.00	−0.00
22	3.34	7.42	3.30	−0.00	−0.00	−0.01
33	3.34	3.30	7.42	−0.00	−0.01	−0.00
23	−0.00	−0.00	−0.00	1.79	−0.00	−0.00
13	−0.00	−0.00	−0.01	−0.00	2.25	0.02
12	−0.00	−0.01	−0.00	−0.00	0.02	2.25

BOUNDARY CONDITIONS

where	boundary condition
this figure (visualization)	Periodic: u(x) = eps_total * E_k . x + u_tilde(x), u_tilde periodic on opposite faces, non-overlapping slave masks per axis (axis 0 excludes y=L,z=L sub-edges; axis 1 excludes z=L; axis 2 takes rest), 3 sub-space pins at the geometric centre
homogenization solver (b3-tex solve –backend periodic)	Same periodic BC, 6 unit Voigt strains; effective stiffness = volume-averaged stress per loadcase
KUBC alternative (b3-tex solve –backend kubc)	u = E_k . x on the entire boundary; gives an upper bound for C_eff

PERIODIC-BC VERIFICATION (machine-precision residuals over all 6 loadcases)

check	worst residual (across all 6 loadcases)
u_tilde slave-master pairs	3.23e-18
u_tilde 8 corners agree	0.00e+00
u_tilde at centre pin	0.00e+00